

Week 1

Expected Work:

Complete basic Python programming exercises covering variables, data types, control flow, functions, collections, and basic OOP, along with regular practice assignments.

Day	Activities Performed / Discussed Topics / Details of Work Allotted
Day 1	Introduction to Python for Data Science and basic programming structure. Discussed Variables, Data Types, and Operators with practical examples. Students practiced basic Python programs and were given exercises for practice.
Day 2	Discussed Control Flow using If-Else statements, including different conditions and decision-making situations. Students practiced conditional programs based on simple real-world problems.
Day 3	Introduced Loops, including for and while loops. Students practiced repetitive tasks, nested loops, and basic loop-based programming problems.
Day 4	Discussed Functions and Modules. Students learned about creating functions, parameters, return values, and using modules. Practical exercises were given based on functions.
Day 5	Covered Python collection types including List, Tuple, Dictionary, and Set. Students practiced creating, accessing, updating, and performing basic operations on these data structures.
Day 6	Introduction to Object-Oriented Programming (OOP). Discussed Classes, Objects, and Constructors with practical examples. Students created simple classes and objects and were given basic OOP exercises.

Week 2

Expected Work:

Complete practical exercises on advanced OOP, exception handling, NumPy, and basic Pandas, including a small dataset-based practice assignment.

Day	Activities Performed / Discussed Topics / Details of Work Allotted
Day 1	Continued Object-Oriented Programming with focus on Inheritance and Encapsulation. Practical examples were discussed and students practiced creating classes using these concepts.
Day 2	Discussed Polymorphism and Abstraction with simple practical examples. Students practiced implementing basic OOP concepts through programming exercises.
Day 3	Introduction to Exception Handling. Discussed common errors and the use of try, except, and related statements. Students practiced handling errors in Python programs.
Day 4	Introduction to NumPy and its importance in Data Science. Covered arrays, array creation, indexing, slicing, and basic array operations through practical exercises.
Day 5	Continued NumPy with mathematical/statistical operations and basic array manipulation. Students practiced working with NumPy arrays and solving data-oriented exercises.
Day 6	Introduction to Pandas and DataFrames. Students learned to create and load DataFrames, inspect data, select columns and rows, and perform basic data operations. A practice assignment was allotted using a sample dataset.

Week 3

Expected Work:

Perform basic data cleaning, preprocessing, visualization, and EDA on a sample dataset using Pandas, Matplotlib, and Seaborn.

Day	Activities Performed / Discussed Topics / Details of Work Allotted
Day 1	Introduction to Data Analysis and Exploratory Data Analysis (EDA). Discussed the importance of understanding a dataset before performing analysis. Students practiced loading and inspecting datasets using Pandas.
Day 2	Covered Data Cleaning concepts including missing values, duplicate records, and basic data inconsistencies. Students practiced identifying and handling these issues using Pandas.
Day 3	Discussed Data Transformation and Preprocessing. Students practiced changing data types, selecting and modifying columns, and performing basic transformations required before analysis.
Day 4	Introduction to Matplotlib for data visualization. Students created basic charts and learned how visualizations can be used to understand patterns and trends in data.
Day 5	Introduction to Seaborn and its use in EDA. Students practiced creating different statistical visualizations and compared the information obtained from different charts.
Day 6	Practical EDA session combining Pandas, Matplotlib, and Seaborn. Students worked on a sample dataset, performed basic cleaning, created visualizations, and discussed the observations obtained from the data.

Week 4

Expected Work:

Complete the individual EDA project by applying data cleaning, feature engineering, scaling, visualization, and analysis techniques, and prepare the project for presentation.

Day	Activities Performed / Discussed Topics / Details of Work Allotted
Day 1	Continued Exploratory Data Analysis with focus on understanding relationships between different variables. Students practiced using visualizations to identify trends, patterns, and unusual observations in the dataset.
Day 2	Discussed Feature Engineering and the importance of creating useful features from existing data. Students practiced basic feature creation and modification using Pandas.
Day 3	Introduction to Data Scaling and Normalization. The need for scaling numerical data was explained, followed by practical implementation using suitable Python libraries.
Day 4	Discussed the complete EDA workflow, including data loading, cleaning, preprocessing, analysis, visualization, and interpretation. Students worked on their selected project datasets.
Day 5	Students worked on their individual EDA projects, applying the concepts covered during the previous sessions. Guidance was provided for data cleaning, visualization, and identifying meaningful insights.
Day 6	Project review and improvement session. Students checked their analysis and visualizations, corrected errors, completed pending work, and prepared their EDA project outputs for presentation.

Week 5

Expected Work:

Create a basic Power BI report using a sample dataset with suitable visualizations and map-based charts.

Day	Activities Performed / Discussed Topics / Details of Work Allotted
Day 1	Introduction to Power BI and Business Intelligence, including its purpose in data analysis and reporting. Discussed the basic Power BI workflow and introduction to Power BI Desktop.
Day 2	Introduced the Power BI Desktop interface, including major sections and basic report-building workflow. Students practiced importing a sample dataset and understanding available fields.
Day 3	Discussed data import and basic data preparation using CSV and Excel files. Students practiced connecting datasets and examining the imported data.
Day 4	Introduced basic Power BI visualizations and their purpose in presenting data. Students created simple charts and learned how fields can be used in different visuals.
Day 5	Continued practical work with charts and visualizations, focusing on selecting suitable visuals according to the type of information being presented.
Day 6	Introduction to Maps and Filled Maps for geographical analysis. Students practiced creating map-based visualizations and were given a basic Power BI assignment covering the concepts learned during the week.

Week 6

Expected Work:

Create an interactive Power BI report using advanced visuals, conditional formatting, filters, and slicers.

Day	Activities Performed / Discussed Topics / Details of Work Allotted
Day 1	Introduction to advanced charts and visualizations in Power BI. Students explored different visual types and discussed their suitable use cases.
Day 2	Practical work on Cards, KPI, Gauge, Table, and Matrix visuals. Students practiced presenting key values and detailed information in reports.
Day 3	Introduction to Tree Map and Funnel Chart along with practical examples. Students created visuals and compared their usefulness for different types of data.
Day 4	Discussed Conditional Formatting in tables and matrices. Students practiced highlighting values based on specific conditions to improve report readability.
Day 5	Introduction to Slicers and Filters for interactive report analysis. Students practiced filtering reports using different fields and values.
Day 6	Practical session combining different Power BI visuals, filters, and slicers to create a more interactive report. Weekly assignment was given based on the concepts covered.

Week 7

Expected Work:

Prepare and transform a sample dataset using Power Query, applying suitable cleaning, transformation, Merge/Append, and basic M Language concepts.

Day	Activities Performed / Discussed Topics / Details of Work Allotted
Day 1	Introduction to Power Query Editor and its role in preparing and transforming data before creating reports in Power BI.
Day 2	Discussed basic data cleaning and transformation techniques such as changing data types, removing unnecessary data, and handling basic data inconsistencies.
Day 3	Practical work on Transformations using Power Query, including renaming columns, removing columns, replacing values, and formatting data.
Day 4	Discussed Merge and Append operations and their basic purpose when working with multiple datasets. Students practiced combining sample data.
Day 5	Introduction to M Language and its role in Power Query. Students were shown how transformations are represented through Power Query steps.
Day 6	Practical session combining different Power Query transformations on a sample dataset. Students completed a weekly assignment based on the data preparation techniques covered.

Week 8

Expected Work:

Create and use calculated columns and basic DAX measures to perform calculations and enhance Power BI reports.

Day	Activities Performed / Discussed Topics / Details of Work Allotted
Day 1	Introduction to DAX (Data Analysis Expressions) and its importance in creating calculations and measures in Power BI.
Day 2	Discussed basic DAX syntax and calculated columns with simple examples. Students practiced creating basic calculated columns from the available dataset.
Day 3	Introduction to commonly used DAX functions for basic calculations and aggregations. Students created simple measures for report analysis.
Day 4	Discussed filter context and basic filtering concepts in DAX through practical examples. Students observed how calculations change based on report selections.
Day 5	Practical work on creating measures and calculated columns and using them with Power BI visuals. Students practiced applying DAX calculations to their reports.
Day 6	Revision and practical session covering the DAX concepts discussed during the week. Students completed a weekly assignment involving basic DAX calculations and measures.

Week 9

Expected Work:

Apply DAX calculations and interactive Power BI features such as Bookmarks and Drillthroughs to develop a more functional analytical report.

Day	Activities Performed / Discussed Topics / Details of Work Allotted
Day 1	Continued DAX with commonly used functions for filtering and working with data in different contexts. Practical examples were used to understand how DAX calculations respond to report selections.
Day 2	Discussed Filter Functions and their practical use in Power BI calculations. Students practiced creating calculations based on selected or filtered data.
Day 3	Practical work on Time and Date-based calculations using suitable DAX functions. Students practiced creating calculations involving dates and analyzing data over time.
Day 4	Introduced Bookmarks and discussed their use in creating interactive report views. Students practiced creating and managing bookmarks within a Power BI report.
Day 5	Introduced Drillthrough functionality and its purpose in providing detailed information from a summary report. Students created a basic drillthrough page and tested navigation between report pages.
Day 6	Practical revision covering DAX, Bookmarks, Drillthrough, Filters, and interactive report features. Students worked on an assignment combining the concepts covered during the week.

Week 10

Expected Work:

Revise and strengthen important Power BI, Power Query, DAX, visualization, and interactive reporting concepts and prepare for the individual Power BI project in Week 11.

Day	Activities Performed / Discussed Topics / Details of Work Allotted
Day 1	Revision of important Power BI concepts covered in the previous weeks, with emphasis on understanding the complete workflow from data import to final report creation.
Day 2	Practical session on Power Query and data preparation, focusing on commonly required transformations and ensuring that data is ready for analysis.
Day 3	Revision and practical application of DAX calculated columns and measures. Students practiced selecting appropriate calculations according to reporting requirements.
Day 4	Advanced practice with visualizations, filters, slicers, and conditional formatting to improve report presentation and readability.
Day 5	Practical work on Bookmarks, Drillthroughs, and report navigation. Students created interactive report features and tested the flow between different report pages.
Day 6	Power BI report design and best practices were discussed, including proper visual selection, page organization, readability, and presenting meaningful information. Students were given preparation work for the individual Power BI project to be undertaken in Week 11.

Week 11

Expected Work:

Complete and finalize an individual Power BI project using a Kaggle dataset, including data preparation, analysis, interactive visualizations, and a multipage report.

Day	Activities Performed / Discussed Topics / Details of Work Allotted
Day 1	Introduction to the individual Power BI project. Project requirements, dataset selection, expected report structure, and overall workflow were explained to the interns.
Day 2	Students selected a relatively large dataset from Kaggle and studied the available columns, data types, and overall purpose of the dataset before starting the report.
Day 3	Students performed data preparation and transformation using Power Query wherever required. Basic data cleaning and organization were carried out before analysis.
Day 4	Students started developing their individual Power BI reports, selecting suitable visualizations and applying the Power BI concepts covered during the training.
Day 5	Students worked on DAX calculations, slicers, filters, bookmarks, drillthroughs, and multipage report design according to the requirements of their individual projects.
Day 6	Project review and correction was conducted. Students checked their reports, corrected errors, improved the report layout, and prepared their final project presentation and demonstration.

Week 12

Expected Work:

Complete the resume, professional grooming activities, project presentation preparation, and basic Generative AI learning, and review the overall learning gained during the internship.

Day	Activities Performed / Discussed Topics / Details of Work Allotted
Day 1	Introduction to Generative AI and discussion on its growing use in the technology and data analytics industry. Basic examples of AI-assisted work were demonstrated.
Day 2	Introduction to AI tools such as ChatGPT and Gemini. Students discussed how such tools can be used for learning, understanding concepts, generating ideas, and resolving basic doubts responsibly.
Day 3	Discussed the importance of professional communication, workplace behaviour, and professional grooming. Students were guided on presenting themselves appropriately in a professional environment.
Day 4	Resume Building session was conducted. Students were guided on preparing a structured resume, highlighting technical skills, projects, education, and relevant internship experience.
Day 5	Students worked on improving their project explanation and presentation skills. Guidance was provided on how to confidently explain the work completed during the internship.
Day 6	Final career guidance and internship review session. Students discussed their learning experience, technical skills gained, project work, and preparation required for future job opportunities.